

ELITE MATHEMATICS

Pearson Edexcel International A Level

Mechanics 1

Adaptive Classified

Chapter 03 · Kinematics Graphs

With Answers

Questions with full worked solutions

- 33 pages
- Past papers up to June 2026

MECHANICS 1 | TOPIC 03

Kinematics Graphs

Your classified route

14

QUESTIONS

150

MARKS

2

FAMILIES

6

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Read the diagram, choose the model, then use the working space to complete the mechanics.

TOPIC 03 | FAMILY 01

Interpret and transform motion graphs

WHAT THIS FAMILY ASKS

Use gradients, signed/unsigned areas and representation changes among s-t, v-t, speed-t and a-t graphs.

VARIANT ROUTE**V01** Read gradients and areas**V02** Convert between graph types**V03** Find an unknown from graph area**9 classified questions**

90 total marks

October 2021 Q6 – complete original question

6.

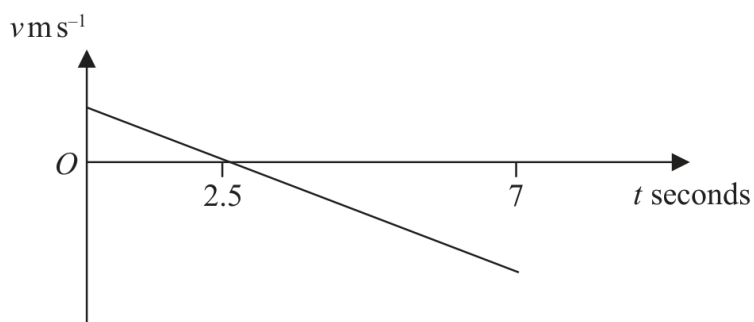


Figure 3

A small ball is thrown vertically upwards at time $t = 0$ from a point A which is above horizontal ground. The ball hits the ground 7 s later.

The ball is modelled as a particle moving freely under gravity.

The velocity-time graph shown in Figure 3 represents the motion of the ball for $0 \leq t \leq 7$

(a) Find the speed with which the ball is thrown.

(2)

(b) Find the height of A above the ground.

(3)

October 2021 Q6 – detailed solution

Read gradients and areas

ALL
5 marks**Use the velocity-time graph**

The velocity becomes zero at $t = 2.5$ s. Since acceleration is $-g = -9.8 \text{ m s}^{-2}$, the initial velocity is $u = 9.8(2.5)$.

Find the initial speed

$$u = 24.5 \text{ m s}^{-1}.$$

Use displacement to the ground

From launch to $t = 7$, the displacement is the area under the velocity-time graph. Algebraically, $s = ut - \frac{1}{2}gt^2$ with $t = 7$.

Calculate displacement

$$s = 24.5(7) - \frac{1}{2}(9.8)(7^2) = 171.5 - 240.1 = -68.6 \text{ m}.$$

State the height

The negative sign shows the ground is below A, so the height of A above the ground is 68.6 m.

Final answer (a) 24.5 m s^{-1} . (b) 68.6 m approximately.

October 2023 Q2 – complete original question

2.

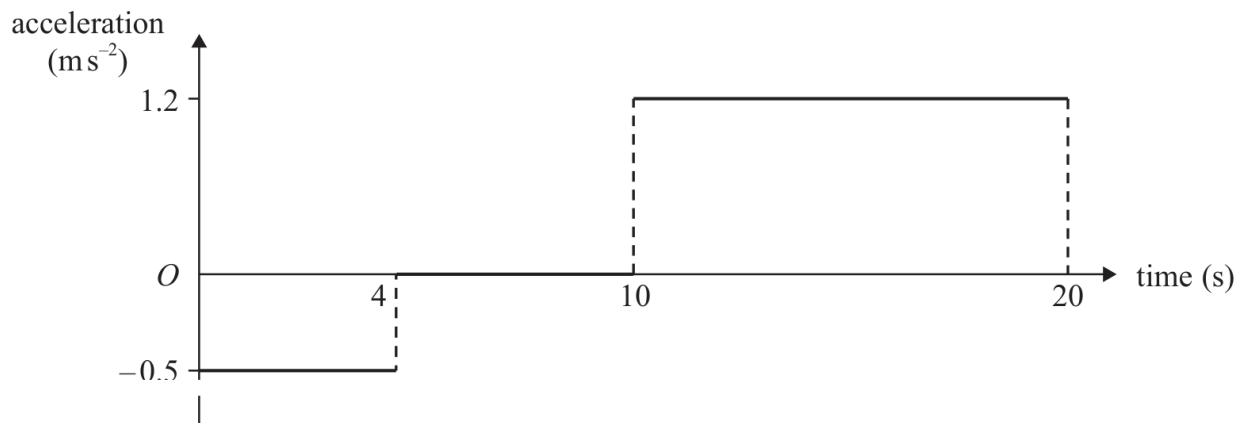


Figure 2

Two fixed points, A and B , are on a straight horizontal road.

The **acceleration-time** graph in Figure 2 represents the motion of a car travelling along the road as it moves from A to B .

At time $t = 0$, the car passes through A with speed 8 m s^{-1}

At time $t = 20 \text{ s}$, the car passes through B with speed $v \text{ m s}^{-1}$

- (a) Show that $v = 18$ (3)
- (b) Sketch a speed-time graph for the motion of the car from A to B . (3)
- (c) Find the distance AB . (4)

October 2023 Q2 – detailed solution

Convert between graph types

ALL
10 marks**Find the speed at $t = 4$**

From 0 to 4 s the acceleration is -0.5 m s^{-2} . Therefore $w = 8 + (-0.5)(4) = 6 \text{ m s}^{-1}$.

Find the final speed

From 10 to 20 s the acceleration is 1.2 m s^{-2} , while the speed at $t = 10$ is still 6. Hence $v = 6 + 1.2(10) = 18 \text{ m s}^{-1}$.

Describe the speed-time graph

The graph starts at (0, 8), falls linearly to (4, 6), stays horizontal to (10, 6), then rises linearly to (20, 18).

Use area under the speed-time graph

The distance is the area under the graph. From 0 to 4: $\frac{8+6}{2} \cdot 4 = 28$. From 4 to 10: $6 \cdot 6 = 36$. From 10 to 20: $\frac{6+18}{2} \cdot 10 = 120$.

Add the distances

$AB = 28 + 36 + 120 = 184 \text{ m}$.

Final answer (a) $v = 18 \text{ m s}^{-1}$. (b) Speed-time graph through (0, 8), (4, 6), (10, 6), (20, 18).
(c) $AB = 184 \text{ m}$.

October 2025 Q6 – complete original question

6.

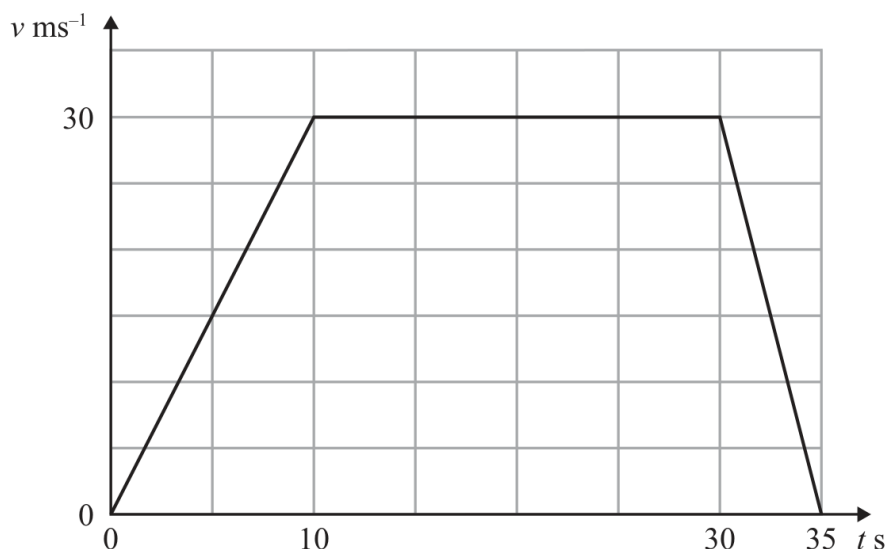


Figure 4

The point O lies on a straight horizontal road.

At time $t = 0$, a car leaves O and travels along the road.

The velocity-time graph in Figure 4 shows the velocity, $v \text{ ms}^{-1}$, of the car at time t seconds for the first 35 seconds of its journey.

(a) Find

(i) the acceleration of the car for the period $0 \leq t \leq 10$

(ii) the deceleration of the car for the period $30 \leq t \leq 35$

(2)

(b) Sketch an acceleration-time graph for the car for the period $0 \leq t \leq 35$

(2)

(c) Find the distance travelled by the car for the period $0 \leq t \leq 35$

(2)

When $t = 5$, a motorcycle starts from rest at O .

The motorcycle travels along the same road as the car and in the same direction.

For the period $5 \leq t \leq 20$, the acceleration of the motorcycle is $A \text{ ms}^{-2}$, where A is a positive constant.

The motorcycle catches up with the car when $t = 20$

(d) Find the value of A .

(4)

October 2025 Q6 – detailed solution

Convert between graph types

ALL
10 marks**Part (a): Find accelerations**(i) Acceleration for $0 \leq t \leq 10$ Acceleration = $\frac{\Delta v}{\Delta t}$ = gradient of v-t graph

$$a = \frac{30-0}{10-0} \quad a = 3 \text{ m s}^{-2}$$

(ii) Deceleration for $30 \leq t \leq 35$

$$a = \frac{0-30}{35-30} \quad a = \frac{-30}{5} \quad a = -6 \text{ m s}^{-2}$$

$$\text{Deceleration} = |-6| = 6 \text{ m s}^{-2}$$

Part (b): Sketch acceleration-time graph

Analysis of each section:

$; 0 \leq t \leq 10$: Constant acceleration = 3 m s^{-2} ; $10 < t \leq 30$: Constant velocity, so acceleration = 0 m s^{-2} ; $30 < t \leq 35$: Constant deceleration = -6 m s^{-2}

Part (c): Find distance travelled for $0 \leq t \leq 35$ - Calculate area under v-t graph

Distance = Area under v-t graph

The shape is a trapezium. We can split it into regions:

Method 1: Trapezium formula

$$\text{Area} = \frac{1}{2}(a+b)h$$

Where the parallel sides are the time intervals and height is velocity.

Better to calculate as sum of shapes:

Method 2: Sum of triangle + rectangle + triangle

$$\text{Area}_1 = \frac{1}{2} \times 10 \times 30 = 150 \text{ m} \quad \text{Area}_2 = 20 \times 30 = 600 \text{ m} \quad \text{Area}_3 = \frac{1}{2} \times 5 \times 30 = 75 \text{ m}$$

$$\text{Total distance} = 150 + 600 + 75 = 825 \text{ m}$$

Part (d): Find value of A - Distance travelled by car (0 to 20 seconds)Area under v-t graph from $t = 0$ to $t = 20$:

$$\text{Distance}_{\text{car}} = \text{Triangle} + \text{Rectangle} = \frac{1}{2}(10)(30) + (10)(30) = 150 + 300 = 450 \text{ m}$$

Part (d): Find value of A - Distance travelled by motorcycle (5 to 20 seconds)Motorcycle starts at $t = 5$ with constant acceleration A .Time of travel: $20 - 5 = 15$ seconds

$$s = ut + \frac{1}{2}at^2$$

$$\text{Distance}_{\text{motorcycle}} = 0 + \frac{1}{2}A(15)^2 = \frac{225A}{2} = 112.5A \text{ m}$$

Part (d): Find value of A - Set distances equalMotorcycle catches up with car when $t = 20$:

$$112.5A = 450 \quad A = \frac{450}{112.5} \quad A = 4 \text{ m s}^{-2}$$

Final answer (a)(i) $a = 3 \text{ m s}^{-2}$, (ii) deceleration = 6 m s^{-2} . (b) correct acceleration-time graph with levels 3, 0, -6. (c) distance = 825 m. (d) $A = 4 \text{ m s}^{-2}$.

January 2026 Q2 – complete original question

2.

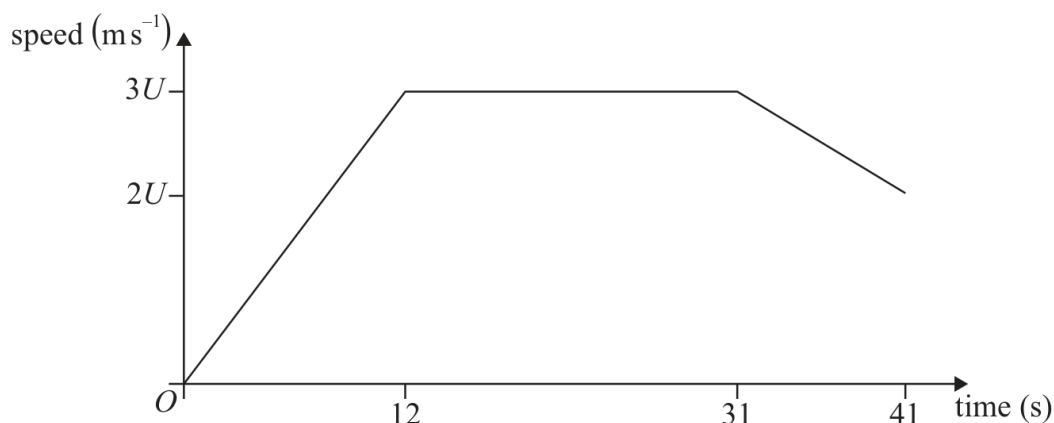


Figure 1

A particle P starts from rest at time $t = 0$ and moves in a straight line

- from $t = 0$ to $t = 12$ s, P accelerates uniformly until it reaches a speed of $3U \text{ m s}^{-1}$
- from $t = 12$ s to $t = 31$ s, P moves with constant speed $3U \text{ m s}^{-1}$
- from $t = 31$ s to $t = 41$ s, P decelerates uniformly until it has speed $2U \text{ m s}^{-1}$

as shown on the speed-time graph in Figure 1.

The distance travelled by P between $t = 0$ and $t = 41$ s is 600 m.

- Show that $U = 6$ (3)
- Find the distance travelled by P whilst moving from rest to a speed of 6 m s^{-1} (2)
- Find the acceleration of P between $t = 0$ and $t = 12$ s. (1)
- Find the deceleration of P between $t = 31$ s and $t = 41$ s. (2)
- Sketch an acceleration-time graph to represent the motion of P from $t = 0$ to $t = 41$ s. (2)

January 2026 Q2 – detailed solution

Convert between graph types

ALL
10 marks**Use area under the speed-time graph**

Distance is the area under a speed-time graph. The three areas are a triangle from 0 to 12, a rectangle from 12 to 31, and a trapezium from 31 to 41.

Show that U is 6

$\frac{1}{2}(12)(3U) + 19(3U) + \frac{1}{2}(3U + 2U)(10) = 600$. This gives $18U + 57U + 25U = 600$, so $100U = 600$ and $U = 6$.

Find the distance to speed 6

Since $U = 6$, the final speed 6 m s^{-1} occurs during the first section. The acceleration there is $\frac{18-0}{12} = 1.5 \text{ m s}^{-2}$, so the time to reach 6 is 4 s. The distance is $\frac{1}{2}(4)(6) = 12 \text{ m}$.

Find the acceleration and deceleration

From 0 to 12, $a = \frac{18}{12} = 1.5 \text{ m s}^{-2}$. From 31 to 41, the speed falls from 18 to 12, so the deceleration is $\frac{18-12}{10} = 0.6 \text{ m s}^{-2}$.

Describe the acceleration-time graph

The acceleration-time graph has three horizontal sections: 1.5 from $0 \leq t \leq 12$, 0 from $12 < t \leq 31$, and -0.6 from $31 < t \leq 41$.

Final answer (a) $U = 6$. (b) 12 m. (c) 1.5 m s^{-2} . (d) 0.6 m s^{-2} . (e) $a(t) = 1.5, 0, -0.6$ on the three intervals.

Summer 2026 Q3 – complete original question

3.

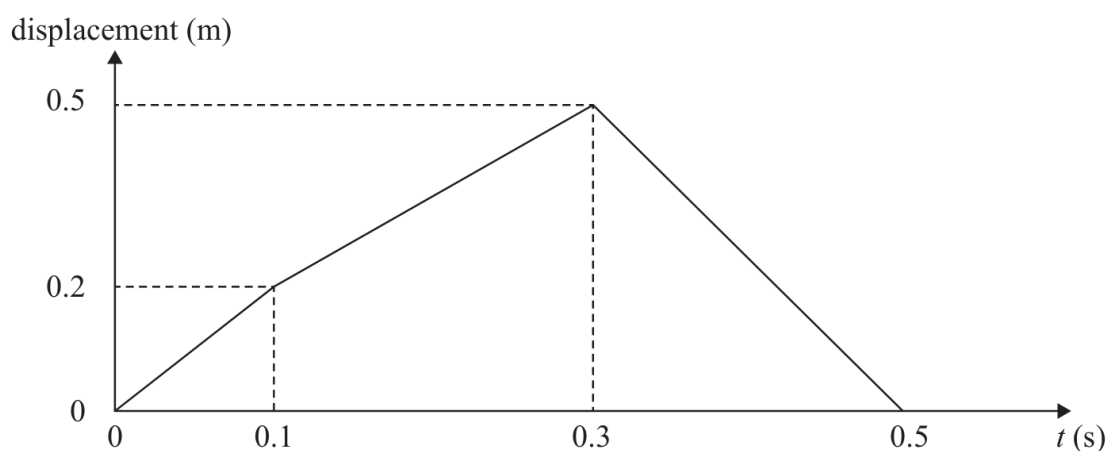


Figure 2

A small ball of mass 0.04 kg moves along a straight line on a pinball table.

At time $t = 0$, the ball passes through the point O

The displacement of the ball is measured from O

Figure 2 shows the **displacement-time** graph for the motion of the ball from $t = 0$ to $t = 0.5$

- (a) State the displacement of the ball from O at $t = 0.5$ (1)
- (b) Find the total distance moved by the ball between $t = 0$ and $t = 0.5$ (1)
- (c) Sketch a velocity-time graph for the motion of the ball from $t = 0$ to $t = 0.5$ (4)
- (d) Find the magnitude of the impulse exerted on the ball at $t = 0.3$, stating the units of your answer. (4)

Summer 2026 Q3 – detailed solution

Convert between graph types

ALL
10 marks**(a) Displacement at 0.5 seconds – Step 1**

At $t = 0.5$ the displacement-time graph returns to the time axis. Therefore the displacement is zero metres.

(b) Total distance travelled – Step 1

Add the absolute displacement changes: $0.2 + 0.3 + 0.5 = 1.0$ metre. The ball travels one metre in total.

(c) Velocity-time graph – Step 1

Velocity is the gradient of each straight section of the displacement-time graph. The gradients are 2, 1.5, and -2.5 metres per second.

(c) Velocity-time graph – Step 2

Draw horizontal velocity segments on $0 \leq t < 0.1$, $0.1 < t < 0.3$, and $0.3 < t \leq 0.5$. Label the three levels 2, 1.5, and -2.5 and the times 0.1, 0.3, and 0.5.

(d) Impulse at the velocity jump – Step 1

Immediately before $t = 0.3$, $v = 1.5$, and immediately after it, $v = -2.5$ metres per second. Using one direction convention, the change in velocity is $-2.5 - 1.5 = -4$ metres per second.

(d) Impulse at the velocity jump – Step 2

The signed impulse is $0.04(-4) = -0.16$ newton seconds. The magnitude is 0.16 newton seconds, equivalent to kilograms metres per second.

Final answer (a) 0 m; (b) 1 m; (c) $v = 2, 1.5, -2.5 \text{ m s}^{-1}$ on the three intervals; (d) 0.16 N s

October 2019 Q6 – complete original question

6. An athlete runs a 200 m race along a straight horizontal track.

In a model of the motion of the athlete, air resistance is ignored, the athlete starts from rest at time $t = 0$ seconds and moves with uniform acceleration 0.8 m s^{-2} for T seconds, reaching a speed of $V \text{ m s}^{-1}$. She then maintains this speed until she crosses the finishing line.

The total time from when the athlete starts to when she crosses the finishing line is 30 s.

- (a) Sketch a speed-time graph for the model of the motion of the athlete from the instant when she starts to the instant when she crosses the finishing line. (2)
- (b) Write down an expression for V in terms of T . (1)
- (c) Show that $T^2 - kT + 500 = 0$, where k is a constant to be found. (4)
- (d) Hence find the value of T , justifying your answer carefully. (3)
- (e) Considering your speed-time graph or otherwise, state two ways, apart from including air resistance, in which the model could be made to be more realistic. (2)

October 2019 Q6 – detailed solution

Find an unknown from graph area

ALL
12 marks**Describe the speed-time graph**

The graph starts at $(0, 0)$, rises in a straight line with gradient 0.8 until time T and speed V , then stays horizontal at speed V until $t = 30$.

Write V in terms of T

Uniform acceleration gives $V = 0.8T$.

Use area for the distance

The distance is the area under the speed-time graph: $\frac{1}{2}TV + V(30 - T) = 200$.

Substitute V

Using $V = 0.8T$: $\frac{1}{2}T(0.8T) + (0.8T)(30 - T) = 200$.

Form the quadratic

$0.4T^2 + 24T - 0.8T^2 = 200$, so $-0.4T^2 + 24T - 200 = 0$. Multiplying by -2.5 gives $T^2 - 60T + 500 = 0$, so $k = 60$.

Solve and choose the valid root

$T^2 - 60T + 500 = (T - 10)(T - 50) = 0$, so $T = 10$ or 50 . Since the whole race lasts 30 s, $T = 10$.

Give two model improvements

A more realistic model could include reaction time at the start, and could replace the instantaneous change from acceleration to constant speed with a smoother transition.

Final answer (a) Speed-time graph rises linearly to V at $t = T$, then is horizontal until $t = 30$. (b) $V = 0.8T$. (c) $T^2 - 60T + 500 = 0$, so $k = 60$. (d) $T = 10$. (e) For example, include reaction time at the start and avoid the instantaneous change to constant speed.

January 2021 Q7 – complete original question

7. A helicopter is hovering at rest above horizontal ground at the point H . A parachutist steps out of the helicopter and immediately falls vertically and freely under gravity from rest for 2.5 s. His parachute then opens and causes him to immediately decelerate at a constant rate of 3.9 m s^{-2} for T seconds ($T < 6$), until his speed is reduced to $V \text{ m s}^{-1}$. He then moves with this constant speed $V \text{ m s}^{-1}$ until he hits the ground. While he is decelerating, he falls a distance of 73.75 m. The total time between the instant when he leaves H and the instant when he hits the ground is 20 s.

The parachutist is modelled as a particle.

- (a) Find the speed of the parachutist at the instant when his parachute opens. (1)
- (b) Sketch a speed-time graph for the motion of the parachutist from the instant when he leaves H to the instant when he hits the ground. (2)
- (c) Find the value of T . (5)
- (d) Find, to the nearest metre, the height of the point H above the ground. (4)

January 2021 Q7 – detailed solution

Find an unknown from graph area

ALL
12 marks**Find speed when parachute opens**

The parachutist falls freely from rest for 2.5 s, so $v = gt = 9.8(2.5) = 24.5 \text{ m s}^{-1}$.

Describe the graph

The speed-time graph starts at 0, rises linearly to 24.5 at $t = 2.5$, then decreases linearly with smaller gradient for T seconds, then stays horizontal at speed V until $t = 20$.

Use the deceleration distance

During the deceleration phase, initial speed is 24.5, acceleration is -3.9 , time is T , and distance is 73.75.

Form the equation for T

$$73.75 = 24.5T + \frac{1}{2}(-3.9)T^2 = 24.5T - 1.95T^2.$$

Choose the valid T

Solving gives $T = 5$ or $T = 7.56 \dots$. Since $T < 6$, take $T = 5$.

Find V

$$V = 24.5 - 3.9(5) = 5 \text{ m s}^{-1}.$$

Find the total height

Height is the total area under the speed-time graph: first triangle $\frac{1}{2}(2.5)(24.5) = 30.625$, deceleration distance 73.75, and final rectangle $(20 - 2.5 - 5)(5) = 62.5$.

Round the height

Total height $= 30.625 + 73.75 + 62.5 = 166.875 \text{ m}$, which rounds to 167 m.

Final answer (a) 24.5 m s^{-1} . (b) Speed-time graph rises to 24.5 at $t = 2.5$, falls to V at $t = 2.5 + T$, then is horizontal to $t = 20$. (c) $T = 5$. (d) $H = 167 \text{ m}$ to the nearest metre.

January 2023 Q1 – complete original question

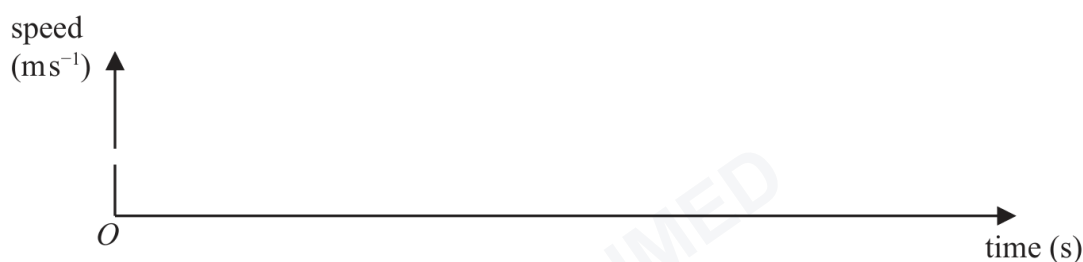
1. A train travels along a straight horizontal track between two stations A and B .

The train starts from rest at station A and accelerates uniformly for T seconds until it reaches a speed of 20 m s^{-1}

The train then travels at a constant speed of 20 m s^{-1} for 3 minutes before decelerating uniformly until it comes to rest at station B .

The magnitude of the acceleration of the train is twice the magnitude of the deceleration.

- (a) On the axes below, sketch a speed–time graph to illustrate the motion of the train as it moves from station A to station B .



If you need to redraw your graph, use the axes on page 3

(3)

Stations A and B are 4.8 km apart.

- (b) Find the value of T

(5)

- (c) Find the acceleration of the train during the first T seconds of its motion.

(2)

Only use these axes if you need to redraw your graph.



(Total for Question 1 is 10 marks)

January 2023 Q1 – detailed solution

Find an unknown from graph area

ALL
10 marks**Relate the acceleration and deceleration times**

The train reaches 20 m s^{-1} in T seconds. Since the acceleration magnitude is twice the deceleration magnitude, the time needed to decelerate from 20 to 0 is $2T$.

Describe the graph

The speed-time graph rises linearly from $(0, 0)$ to $(T, 20)$, stays horizontal at speed 20 from T to $T + 180$, then falls linearly to zero at $3T + 180$.

Use area under the graph

The distance from A to B is $4.8 \text{ km} = 4800 \text{ m}$. The area is $\frac{1}{2}(T)(20) + 20(180) + \frac{1}{2}(2T)(20)$.

Solve for T

$4800 = 10T + 3600 + 20T$, so $1200 = 30T$ and $T = 40 \text{ s}$.

Find the acceleration

During the first T seconds, $20 = aT$. With $T = 40$, $a = \frac{20}{40} = 0.5 \text{ m s}^{-2}$.

Final answer (a) Speed-time graph with times T , $T + 180$, $3T + 180$ and top speed 20. (b) $T = 40 \text{ s}$. (c) $a = 0.5 \text{ m s}^{-2}$.

January 2025 Q2 – complete original question

2. The fixed points A , B and C lie in a straight line on a horizontal road.
- At time $t = 0$, a motorbike passes through A with speed 5 m s^{-1}
 - From A , the motorbike accelerates uniformly until it reaches B with a speed of $V \text{ m s}^{-1}$
 - The motorbike takes T_1 seconds to travel from A to B
 - From B , the motorbike decelerates uniformly until it comes to rest at C
 - The motorbike takes T_2 seconds to travel from B to C
- (a) Sketch a speed-time graph for the motion of the motorbike as it moves from A to C . (3)

The distance AB is 132 m and the distance BC is 136 m.

- (b) Find, in terms of V , an expression for

(i) T_1

(ii) T_2

(4)

Given that the motorbike takes 28 s to travel from A to C ,

- (c) find the value of V ,

(2)

- (d) find the deceleration of the motorbike.

(2)

January 2025 Q2 – detailed solution

Find an unknown from graph area

ALL
11 marks**Sketch the speed-time graph**

The graph starts at speed 5 when $t = 0$, rises linearly to speed V at time T_1 , then falls linearly to speed 0 at time $T_1 + T_2$.

Use area from A to B

The area under the first section is the distance $AB = 132$. Thus $\frac{1}{2}(5 + V)T_1 = 132$, so $T_1 = \frac{264}{V + 5}$.

Use area from B to C

The second section is a triangle of height V and base T_2 . Since $BC = 136$, $\frac{1}{2}VT_2 = 136$, so $T_2 = \frac{272}{V}$.

Use the total time

The whole journey takes 28 s, so $\frac{264}{V + 5} + \frac{272}{V} = 28$.

Solve for V

Multiplying by $V(V + 5)$ gives $264V + 272(V + 5) = 28V(V + 5)$. This simplifies to $7V^2 - 99V - 340 = 0$, so $(V - 17)(7V + 20) = 0$. Since speed is positive, $V = 17$.

Find the deceleration

For the final section, $T_2 = \frac{272}{17} = 16$ s. The speed falls from 17 to 0, so the deceleration is $\frac{17}{16} = 1.0625 \text{ m s}^{-2}$.

Final answer (b)(i) $T_1 = \frac{264}{V + 5}$. (b)(ii) $T_2 = \frac{272}{V}$. (c) $V = 17$. (d) deceleration $= \frac{17}{16} = 1.0625 \text{ m s}^{-2}$.

TOPIC 03 | FAMILY 02

Two-mover graph comparison and chase

WHAT THIS FAMILY ASKS

Compare distances/displacements of two movers represented on a shared or staggered time axis.

VARIANT ROUTE**V01** Staggered start**V02** Overtake or catch-up**V03** Two profiles constrained to finish together**5 classified questions**

60 total marks

Summer 2021 Q8 – complete original question

8. Two trams, tram A and tram B , run on parallel straight horizontal tracks. Initially the two trams are at rest in the depot and level with each other.

At time $t = 0$, tram A starts to move. Tram A moves with constant acceleration 2 m s^{-2} for 5 seconds and then continues to move along the track at constant speed.

At time $t = 20$ seconds, tram B starts from rest and moves in the same direction as tram A . Tram B moves with constant acceleration 3 m s^{-2} for 4 seconds and then continues to move along the track at constant speed.

The trams are modelled as particles.

- (a) Sketch, on the same axes, a speed-time graph for the motion of tram A and a speed-time graph for the motion of tram B , from $t = 0$ to the instant when tram B overtakes tram A .

(3)

At the instant when the two trams are moving with the same speed, tram A is d metres in front of tram B .

- (b) Find the value of d .

(5)

- (c) Find the distance of the trams from the depot at the instant when tram B overtakes tram A .

(5)

(Total 13 marks)

Summer 2021 Q8 – detailed solution

Staggered start

ALL
13 marks**Describe tram A graph**

Tram A starts at $t = 0$, accelerates at 2 m s^{-2} for 5 s, reaching speed 10 m s^{-1} , then continues at this constant speed.

Describe tram B graph

Tram B starts at $t = 20$, accelerates at 3 m s^{-2} for 4 s, reaching speed 12 m s^{-1} , then continues at this constant speed. Its graph crosses A's distance later.

Find when speeds are equal

Tram A's constant speed is 10. Tram B reaches speed 10 after $\frac{10}{3}$ s of acceleration, so the time from $t = 0$ is $20 + \frac{10}{3} = \frac{70}{3}$ s.

Distance of A then

A travels $\frac{1}{2}(5)(10) = 25$ m in its acceleration phase, then $10\left(\frac{70}{3} - 5\right) = \frac{550}{3}$ m. Total = $\frac{625}{3}$ m.

Distance of B then

B is still accelerating from rest, so its distance is $\frac{1}{2}\left(\frac{10}{3}\right)(10) = \frac{50}{3}$ m.

Find d

$$d = \frac{625}{3} - \frac{50}{3} = \frac{575}{3} \text{ m.}$$

Set overtake distances equal

For $t > 24$, A's distance is $25 + 10(t - 5) = 10t - 25$. B's distance is $\frac{1}{2}(4)(12) + 12(t - 24) = 12t - 264$.

Find the overtake time

Set $10t - 25 = 12t - 264$. Then $2t = 239$ and $t = 119.5$ s.

Find the distance from the depot

Using A's distance, $10(119.5) - 25 = 1170$ m.

Final answer (a) Tram A reaches speed 10 at $t = 5$; tram B starts at $t = 20$, reaches speed 12 at $t = 24$, then later overtakes A. (b) $d = \frac{575}{3} \text{ m} \approx 192 \text{ m}$. (c) 1170 m.

Summer 2023 Q5 – complete original question

5.

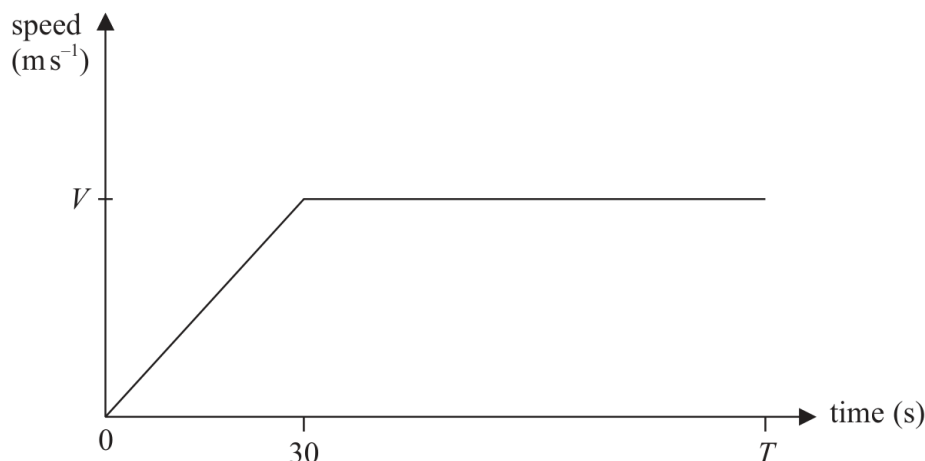


Figure 2

The speed-time graph in Figure 2 illustrates the motion of a car travelling along a straight horizontal road.

At time $t = 0$, the car starts from rest and accelerates uniformly for 30 s until it reaches a speed of $V \text{ ms}^{-1}$.

The car then travels at a constant speed of $V \text{ ms}^{-1}$ until time $t = T$ seconds.

- (a) Show that the distance travelled by the car between $t = 0$ and $t = T$ seconds is $V(T - 15)$ metres.

(2)

A motorbike also travels along the same road.

- The motorbike starts from rest at time $t = 10 \text{ s}$ and accelerates uniformly for 40 s
- The acceleration of the motorbike is the **same** as the acceleration of the car
- The motorbike then travels at a constant speed for a further 10 s before decelerating uniformly until it reaches a speed of $V \text{ ms}^{-1}$ at time T seconds

- (b) On Figure 2, sketch a speed-time graph for the motion of the motorbike.

[If you need to redraw your sketch, there is a copy of Figure 2 on page 15.]

(2)

- (c) Show that the constant speed of the motorbike is $\frac{4V}{3} \text{ ms}^{-1}$

(2)

At time $t = T$ seconds, the distance travelled by each vehicle is the same.

- (d) Find the value of T

(5)

Summer 2023 Q5 – original question continued

Only use this copy of Figure 2 if you need to redraw your sketch.

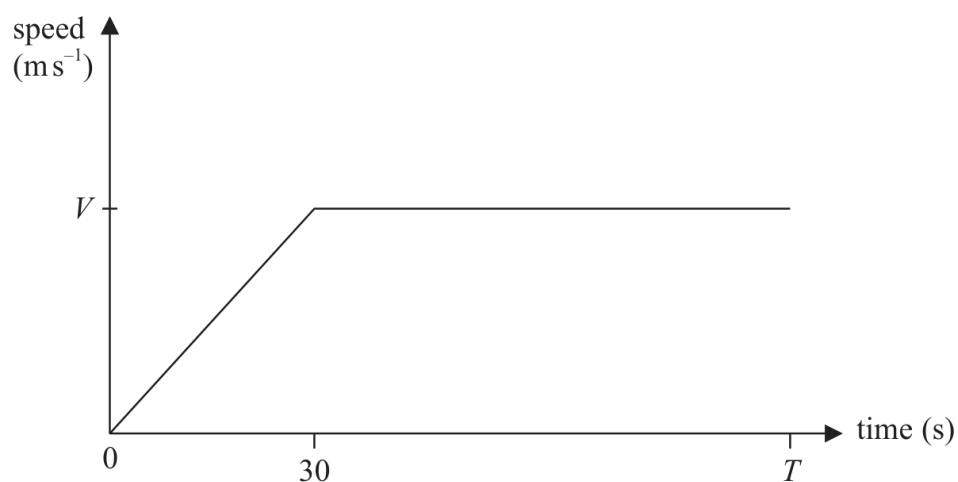


Figure 2

(Total for Question 5 is 11 marks)

Summer 2023 Q5 – detailed solution

Staggered start

ALL
11 marks**Find the car distance**

The car graph is a triangle from 0 to 30 plus a rectangle from 30 to T . Distance = $\frac{1}{2}(30)V + V(T - 30)$.

Simplify the car distance

$\frac{1}{2}(30)V + V(T - 30) = 15V + VT - 30V = V(T - 15)$, as required.

Describe the motorbike graph

The motorbike starts at $(10, 0)$, accelerates with the same gradient as the car until $t = 50$, then travels at constant speed until $t = 60$, then decelerates linearly to (T, V) .

Find the motorbike constant speed

The car acceleration is $\frac{V}{30}$. The motorbike accelerates for 40 s, so its speed is $40 \cdot \frac{V}{30} = \frac{4V}{3}$.

Find the motorbike distance

Its distance is $\frac{1}{2}(40) \left(\frac{4V}{3} \right) + 10 \left(\frac{4V}{3} \right) + \frac{1}{2} \left(\frac{4V}{3} + V \right) (T - 60)$.

Simplify the motorbike distance

This becomes $\frac{80V}{3} + \frac{40V}{3} + \frac{7V}{6}(T - 60) = 40V + \frac{7V}{6}(T - 60) = \frac{7VT}{6} - 30V$.

Equate the two distances

At time T , the distances are equal: $V(T - 15) = \frac{7VT}{6} - 30V$. Divide by V to get $T - 15 = \frac{7T}{6} - 30$.

Find T

Multiplying by 6 gives $6T - 90 = 7T - 180$, so $T = 90$.

Final answer (a) car distance = $V(T - 15)$. (b) Motorbike graph starts at $t = 10$, reaches $\frac{4V}{3}$ at $t = 50$, stays constant to $t = 60$, then falls to V at $t = T$. (c) constant speed = $\frac{4V}{3}$. (d) $T = 90$.

October 2024 Q4 – complete original question

4. The points A and B lie on the same straight horizontal road.

Figure 2, on page 11, shows the speed-time graph of a cyclist P , for his journey from A to B .

At time $t = 0$, P starts from rest at A and accelerates uniformly for 9 seconds until his speed is $V \text{ m s}^{-1}$

He then travels at constant speed $V \text{ m s}^{-1}$

When $t = 42$, cyclist P passes B .

Given that the distance AB is 120 m,

- (a) show that $V = 3.2$ (3)

- (b) Find the acceleration of cyclist P between $t = 0$ and $t = 9$ (2)

Cyclist P continues to cycle along the road in the same direction at the same constant speed, $V \text{ m s}^{-1}$

When $t = 6$, a second cyclist Q sets off from A and travels in the same direction as P along the same road. She accelerates for T seconds until her speed is 3.6 m s^{-1}

She then travels at constant speed 3.6 m s^{-1}

Cyclist Q catches up with P when $t = 54$

- (c) On Figure 2, on page 11, sketch a speed-time graph showing the journeys of **both** cyclists, for the interval $0 \leq t \leq 54$ (3)

- (d) Find the value of T (5)

October 2024 Q4 – original question continued

Question 4 continued

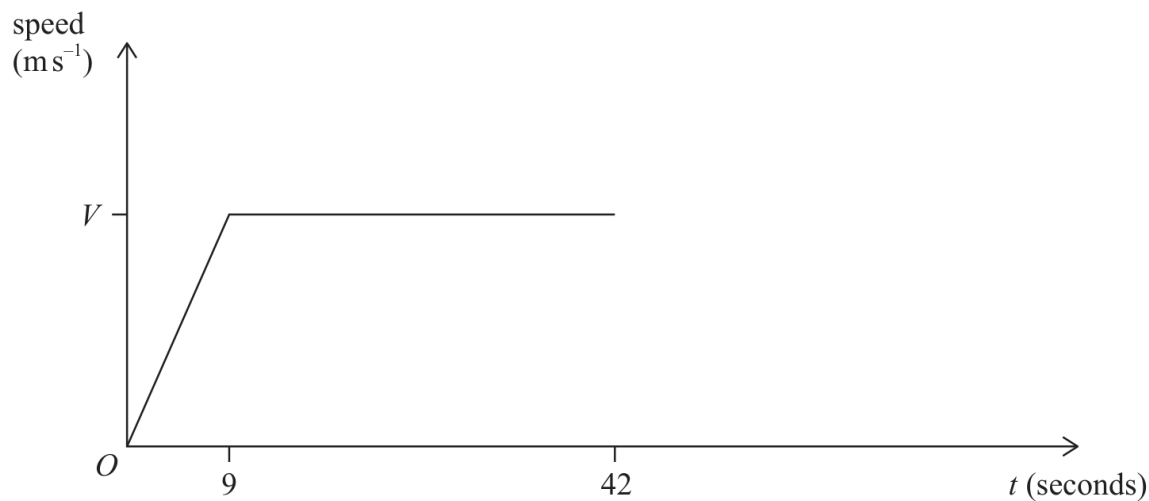
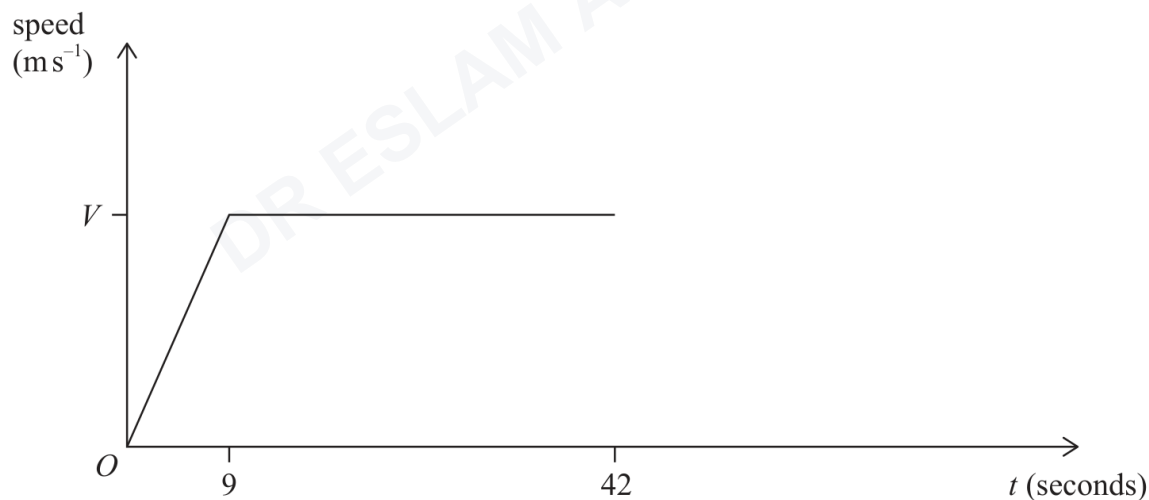


Figure 2

A copy of Figure 2 is on page 13 if you need to redraw your answer to part (c).

Question 4 continued

Only use this copy of Figure 2 if you need to redraw your answer to part (c).



Copy of Figure 2

October 2024 Q4 – detailed solution

Staggered start

ALL
13 marks**Use the area under P graph**

For cyclist P, the distance from A to B is the area under the speed-time graph from 0 to 42. This is $\frac{1}{2}(9)V + 33V = 120$.

Show V is 3.2

$$4.5V + 33V = 37.5V = 120, \text{ so } V = 3.2 \text{ m s}^{-1}.$$

Find P acceleration

Between $t = 0$ and $t = 9$, P accelerates from 0 to 3.2. Hence $a = \frac{3.2}{9} = \frac{16}{45} = 0.356 \dots \text{ m s}^{-2}$.

Describe Q graph

Cyclist Q starts at $t = 6$ from rest, accelerates in a straight line for T seconds to speed 3.6 m s^{-1} , then continues horizontally at 3.6 m s^{-1} until $t = 54$.

Find P distance by t equals 54

By $t = 54$, P has travelled $\frac{1}{2}(9)(3.2) + 45(3.2) = 14.4 + 144 = 158.4 \text{ m}$.

Form Q distance in terms of T

Q travels for 48 seconds after starting at $t = 6$. Her distance is $\frac{1}{2}T(3.6) + (48 - T)(3.6) = 172.8 - 1.8T$.

Solve for T

Q catches P at $t = 54$, so $172.8 - 1.8T = 158.4$. Thus $1.8T = 14.4$ and $T = 8 \text{ s}$.

Final answer (a) $V = 3.2$. (b) $a = \frac{16}{45} \text{ m s}^{-2} \approx 0.36 \text{ m s}^{-2}$. (d) $T = 8 \text{ s}$.

January 2020 Q5 – complete original question

5. A car travels at a constant speed of 40 m s^{-1} in a straight line along a horizontal racetrack. At time $t = 0$, the car passes a motorcyclist who is at rest. The motorcyclist immediately sets off to catch up with the car.

The motorcyclist accelerates at 4 m s^{-2} for 15 s and then accelerates at 1 m s^{-2} for a further T seconds until he catches up with the car.

- (a) Sketch, on the same axes, the speed-time graph for the motion of the car and the speed-time graph for the motion of the motorcyclist, from time $t = 0$ to the instant when the motorcyclist catches up with the car.

(2)

At the instant when $t = t_1$ seconds, the car and the motorcyclist are moving at the same speed.

- (b) Find the value of t_1

(2)

- (c) Show that $T^2 + kT - 300 = 0$, where k is a constant to be found.

(6)

January 2020 Q5 – detailed solution

Overtake or catch-up

ALL
10 marks**Describe the graph**

The car's graph is a horizontal line at 40 m s^{-1} . The motorcyclist's graph starts at zero, rises with gradient 4 until $t = 15$, then rises with smaller gradient 1 for a further T seconds.

Find the equal-speed time

Before $t = 15$, the motorcyclist's speed is $4t$. Set this equal to the car's speed: $4t_1 = 40$, so $t_1 = 10 \text{ s}$.

Find the motorcyclist's speed at 15 seconds

After the first 15 seconds, the motorcyclist's speed is $4(15) = 60 \text{ m s}^{-1}$.

Find the final speed at catch-up

During the next T seconds the acceleration is 1 m s^{-2} , so the speed at catch-up is $60 + T$.

Equate distances

Motorcyclist distance = $\frac{1}{2}(15)(60) + \frac{1}{2}[60 + (60 + T)]T$. Car distance = $40(15 + T)$.

Simplify the equation

$450 + 60T + \frac{1}{2}T^2 = 600 + 40T$. Hence $\frac{1}{2}T^2 + 20T - 150 = 0$, so $T^2 + 40T - 300 = 0$ and $k = 40$.

Final answer (a) Car: horizontal line at 40; motorcyclist: two rising straight-line sections ending at the catch-up time. (b) $t_1 = 10 \text{ s}$. (c) $T^2 + 40T - 300 = 0$, so $k = 40$.

Summer 2022 Q7 – complete original question

7. Two small children, Ajaz and Beth, are running a 100 m race along a straight horizontal track.

They both start from rest, leaving the start line at the same time.

Ajaz accelerates at 0.8 ms^{-2} up to a speed of 4 ms^{-1} and then maintains this speed until he crosses the finish line.

Beth accelerates at 1 ms^{-2} for T seconds and then maintains a constant speed until she crosses the finish line.

Ajaz and Beth cross the finish line at the same time.

- (a) Sketch, on the same axes, a speed-time graph for each child, from the instant when they leave the start line to the instant when they cross the finish line. (3)
- (b) Find the time taken by Ajaz to complete the race. (4)
- (c) Find the value of T (4)
- (d) Find the difference in the speeds of the two children as they cross the finish line. (2)

Summer 2022 Q7 – detailed solution

Two profiles constrained to finish together

ALL
13 marks**Describe Ajaz graph**

Ajaz starts from rest, accelerates linearly to speed 4 m s^{-1} , then continues with a horizontal speed-time line at speed 4.

Describe Beth graph

Beth starts from rest, has a steeper straight acceleration line for T seconds, then a horizontal line. Both graphs finish at the same final time.

Find Ajaz acceleration time

Ajaz reaches speed 4 with acceleration 0.8, so the time is $\frac{4}{0.8} = 5 \text{ s}$.

Find Ajaz total time

Distance during acceleration is $\frac{1}{2}(5)(4) = 10 \text{ m}$. The remaining 90 m at speed 4 takes 22.5 s, so Ajaz's total time is 27.5 s.

Set Beth distance equation

Beth accelerates for T seconds to speed T , then runs at speed T until time 27.5. Her distance is $\frac{1}{2}T^2 + T(27.5 - T)$.

Solve for T

$100 = \frac{1}{2}T^2 + 27.5T - T^2 = 27.5T - \frac{1}{2}T^2$. Thus $T^2 - 55T + 200 = 0$. The valid root is $T = 3.915 \dots \text{ s}$.

Find the finish speeds

Ajaz finishes at 4 m s^{-1} . Beth's constant speed is $T = 3.915 \dots \text{ m s}^{-1}$.

Find the difference

Difference = $4 - 3.915 \dots = 0.085 \text{ m s}^{-1}$ approximately.

Final answer (a) Ajaz reaches speed 4 then runs horizontally; Beth has a steeper initial line to speed T , then horizontal, both ending at the same time. (b) 27.5 s. (c) $T = 3.92 \text{ s}$ approximately. (d) 0.085 m s^{-1} approximately.